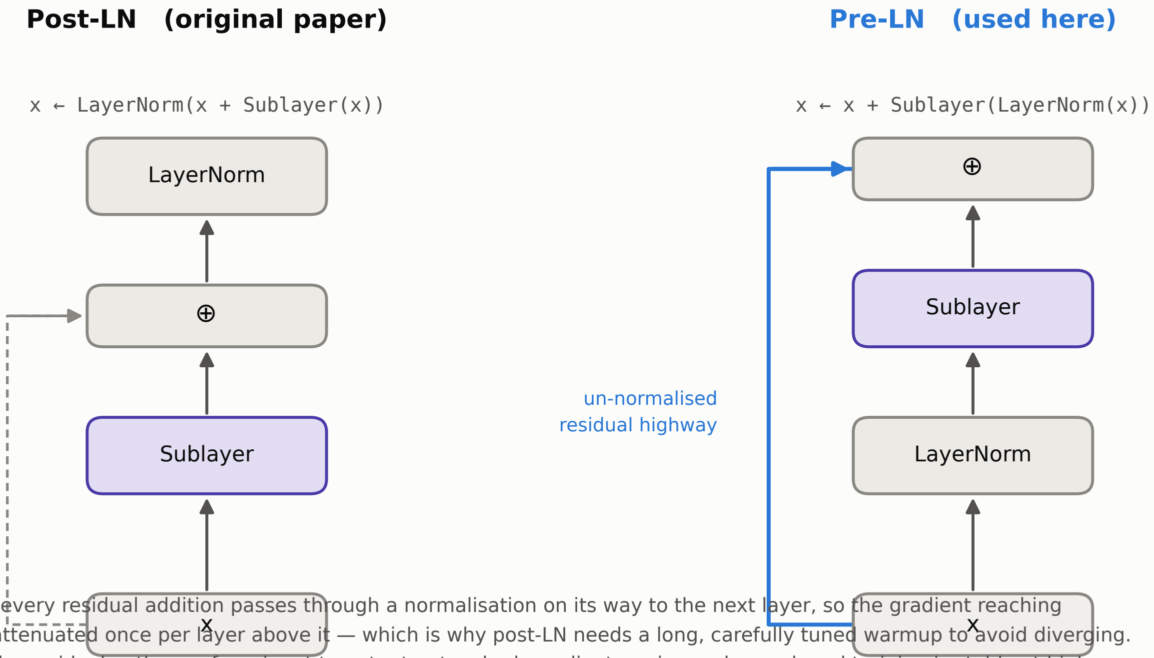
Where to put the layer normalisation



In post-LN every residual addition passes through a normalisation on its way to the next layer, so the gradient reaching layer 1 is attenuated once per layer above it — which is why post-LN needs a long, carefully tuned warmup to avoid diverging. In pre-LN the residual path runs from input to output untouched, gradients arrive undamped, and training is stable at higher learning rates with a shorter warmup. That is why this project uses it: a Colab run that diverges at step 300 is an hour lost.